DATA 230 Group Project Mid-Presentation

Presented by:

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Dataset Introduction and Motivation

Dataset Source

Santa Clara Valley Transportation Authority (VTA)

data.vta.org/pages/ridership-by-stop

Dataset Size

382,720 records with **35 features** (columns)

Each record represents ridership activity at a transit stop

Dataset Type

Public Transit Ridership (RBS Dataset)

Stop-level ridership data with geographic coordinates

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Key Features

Route Information

Route Name and Route Number for identifying specific transit lines

Service Characteristics

Service Period (Weekday / Weekend) for temporal analysis

Location Details

Stop ID, City, Direction for geographic context

Ridership Metrics

Boardings, Alightings, Total Ridership for usage patterns

Spatial Data

Geographic Coordinates for mapping and visualization

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How the Dataset Aligns with Project Goals

Real-World Public Transit Data

Provides actual ridership data from Santa Clara County

Urban Mobility Analysis

Highly relevant for analyzing urban mobility patterns and transportation usage

Pattern Identification

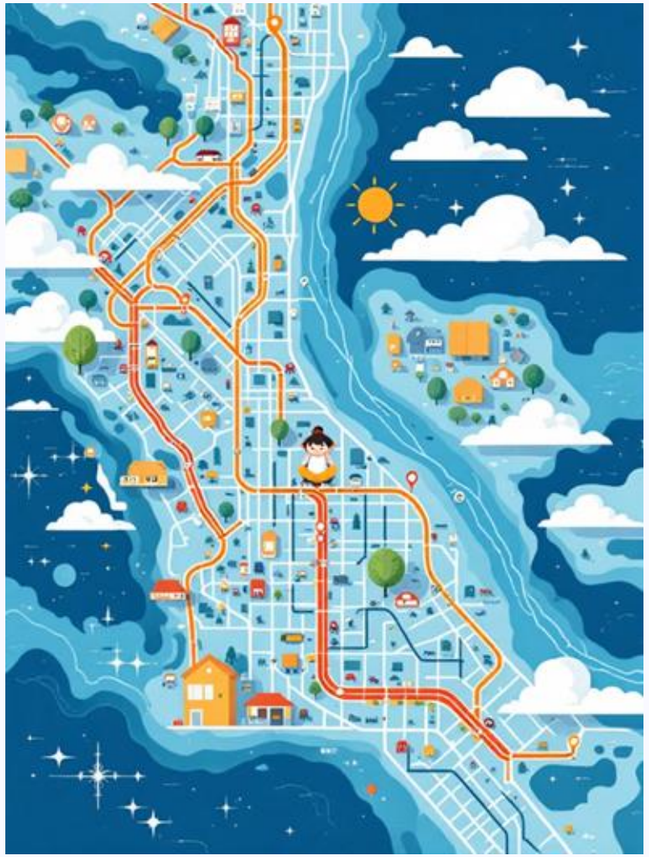
Helps identify high-demand routes, peak usage periods, and geographic trends

Data-Driven Decision Making

Supports data-driven decision making for transit planning and optimization

Exploratory Analysis

Aligns with project goals of performing exploratory data analysis and visualization



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Dataset Complexity

Large-Scale Dataset

Over **380,000 records**, requiring efficient data handling and processing techniques

Data Quality Issues

Contains **missing (null) values** and inconsistent entries that require cleaning and validation

Data Type Mix

Mix of **categorical data** (route name, service period, city) and **numerical data** (boardings, alightings, ridership)

High Granularity

Highly **granular (stop-level data)** with multiple records per route and time period

Spatial Complexity

Includes **geographic information**, enabling spatial analysis but increasing complexity

Multi-Dimensional Analysis

Rich dataset supports **multi-dimensional analysis** across time, location, and ridership patterns

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Data Cleaning & Processing

Drop Empty Data
Removed fully empty rows and columns

Fix Numeric Columns
Coerced 13 columns (BOARDINGS, ALIGHTINGS, PEAK_LOAD, TRIPS, etc.) to numeric. Invalid entries converted to NaN

Standardize String Columns
Stripped whitespace and normalized 12 text columns (CITY, ROUTE_NAME, SERVICE_CODE, etc.). Replaced "nan", "None", and empty strings with proper null values

Convert TRIP_TIME
Original data stored time as Excel day-fractions (e.g., 0.75 = 6:00 PM). Converted to TRIP_HOUR (0-23) for time-based analysis

Fill Missing Values
CITY → "Unknown". BOARDINGS, ALIGHTINGS, and averages → 0

Derive New Features
ROUTE_TYPE: Classified from SERVICE_CODE into 6 categories: Express, Frequent, Local, Light Rail, Rapid, Other. TIME_PERIOD: AM Peak (5-9), Midday (9-3), PM Peak (3-7), Evening (7-11), Late Night (11-5). TOTAL_ACTIVITY: BOARDINGS + ALIGHTINGS

Remove Duplicates
Identified and removed duplicate records

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Exploratory Data Analysis (EDA)

Visualizations Created

- Bar charts for each variable to visualize overall distributions and compare values across the dataset
- Correlation heatmap to examine relationships between variables and identify potential associations among routes
- Box plot to highlight variability in ridership, including differences between minimum, median, and maximum values across routes
- Pie chart to illustrate the proportion of total ridership attributed to each route
- Bar charts of route names to display the top 10 routes by ridership, emphasizing the most utilized routes
- Scatter plot with color intensity (heat-style visualization) to show the relationship between route numbers and ridership levels for enhanced visual interpretation

EDA Insights

Top Route

The **22 Palo Alto - Eastridge** route has the most ridership of any distinguished routes on the VTA routes

Data Variation

The box plots show that the data between columns displays variation in data ranges

Univariate Analysis

The distribution of numeric variables shows a univariate analysis of the data

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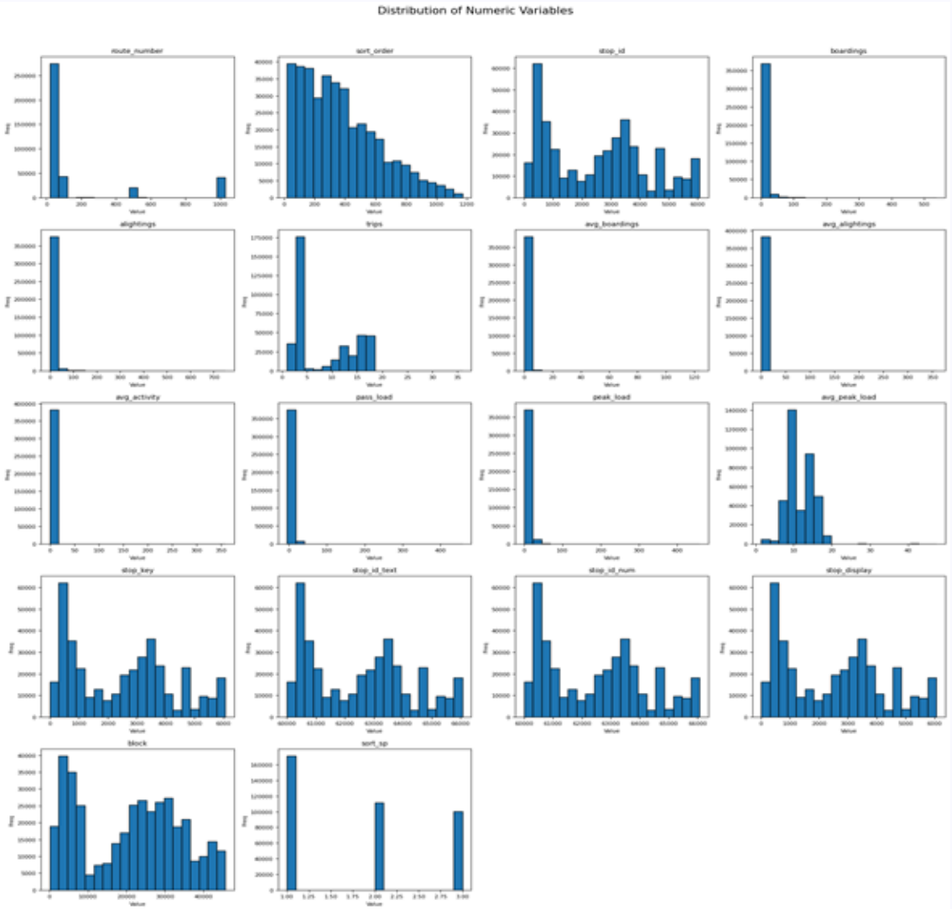
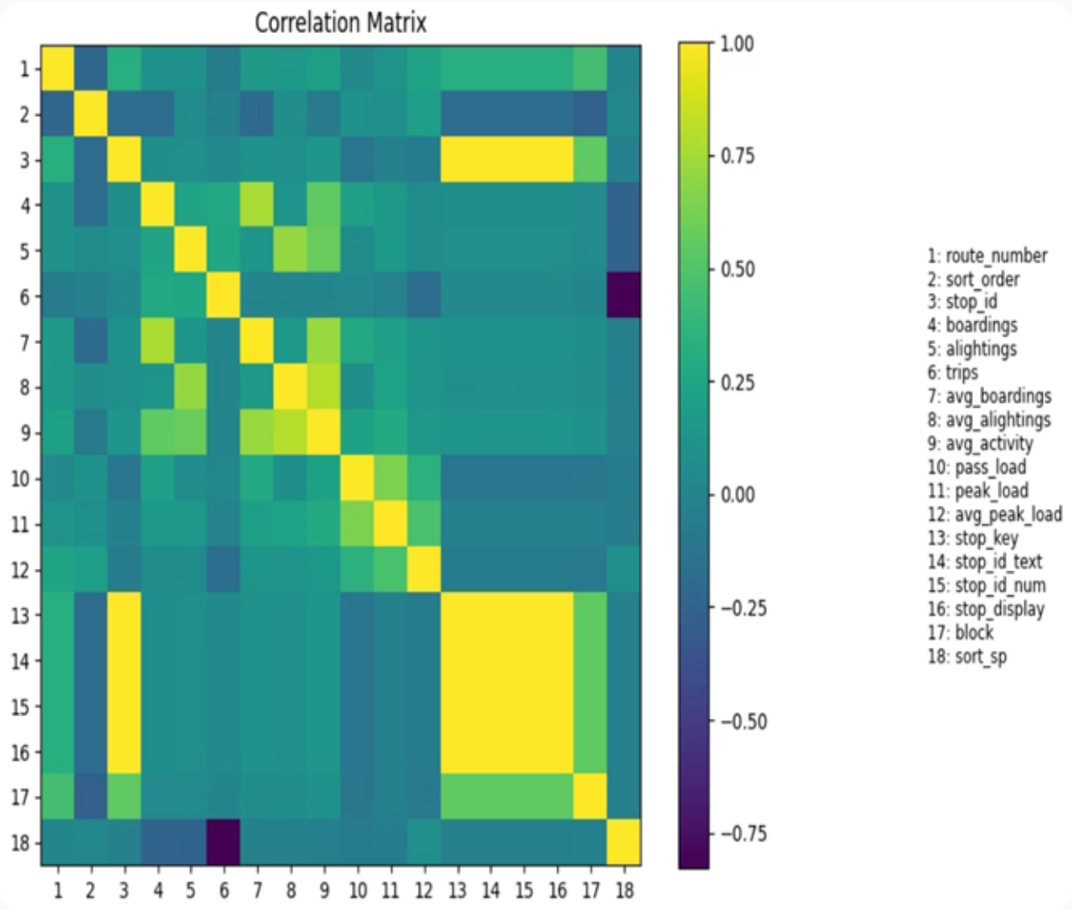
Exploratory Data Analysis (EDA) part 1



Route Ridership Analysis

Top routes by ridership with detailed breakdown



Correlation Heatmap

Variable relationships and route associations

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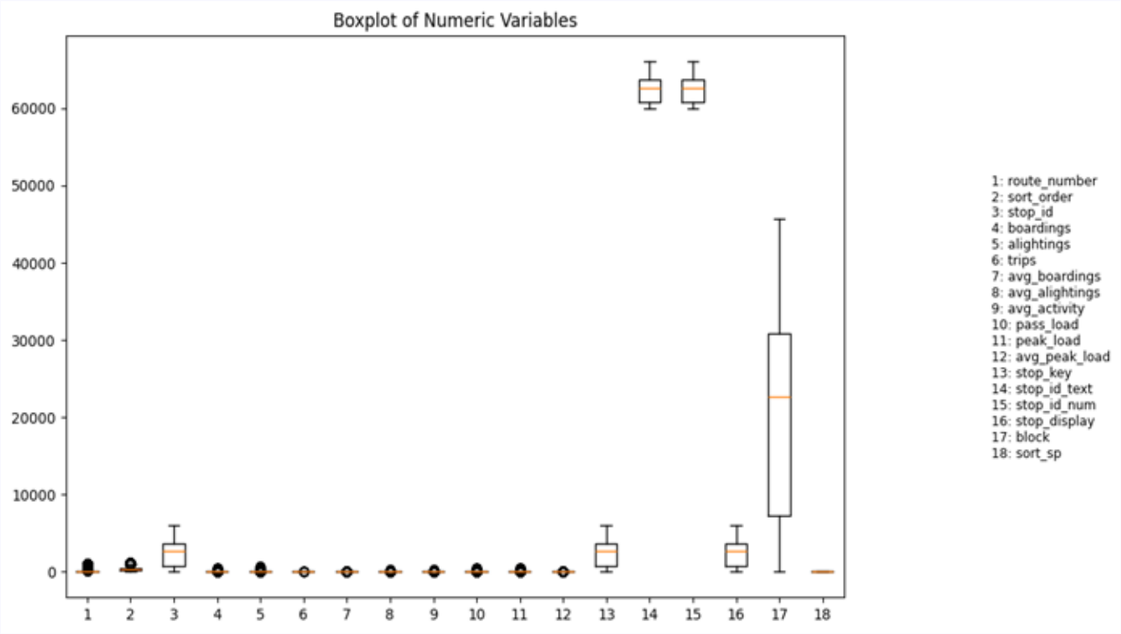
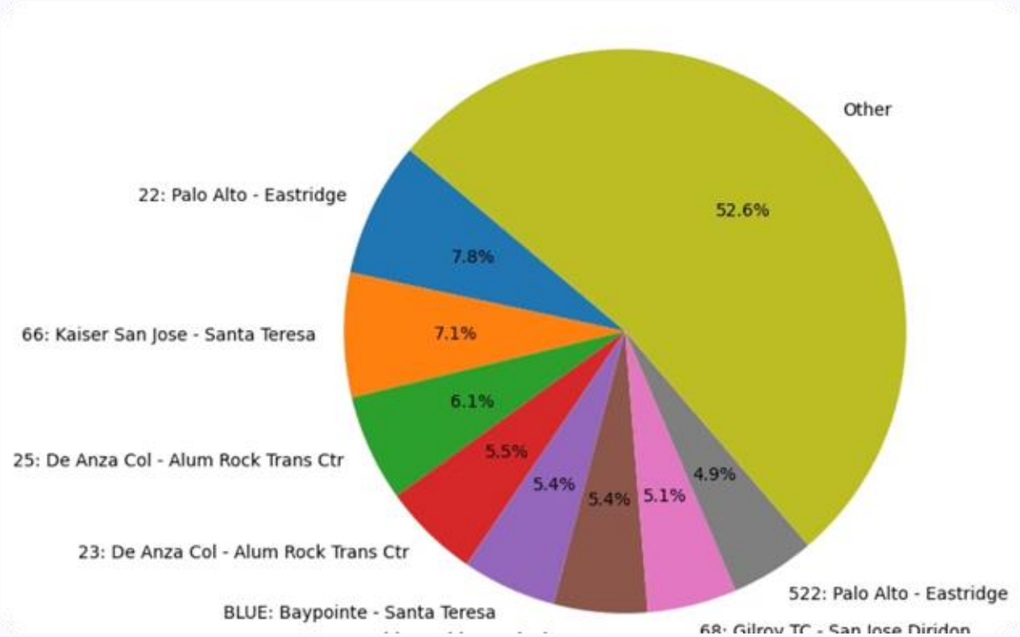
Exploratory Data Analysis (EDA) part 2



Box Plot Analysis

Variability in ridership across routes



Ridership Distribution

Proportion of total ridership by route

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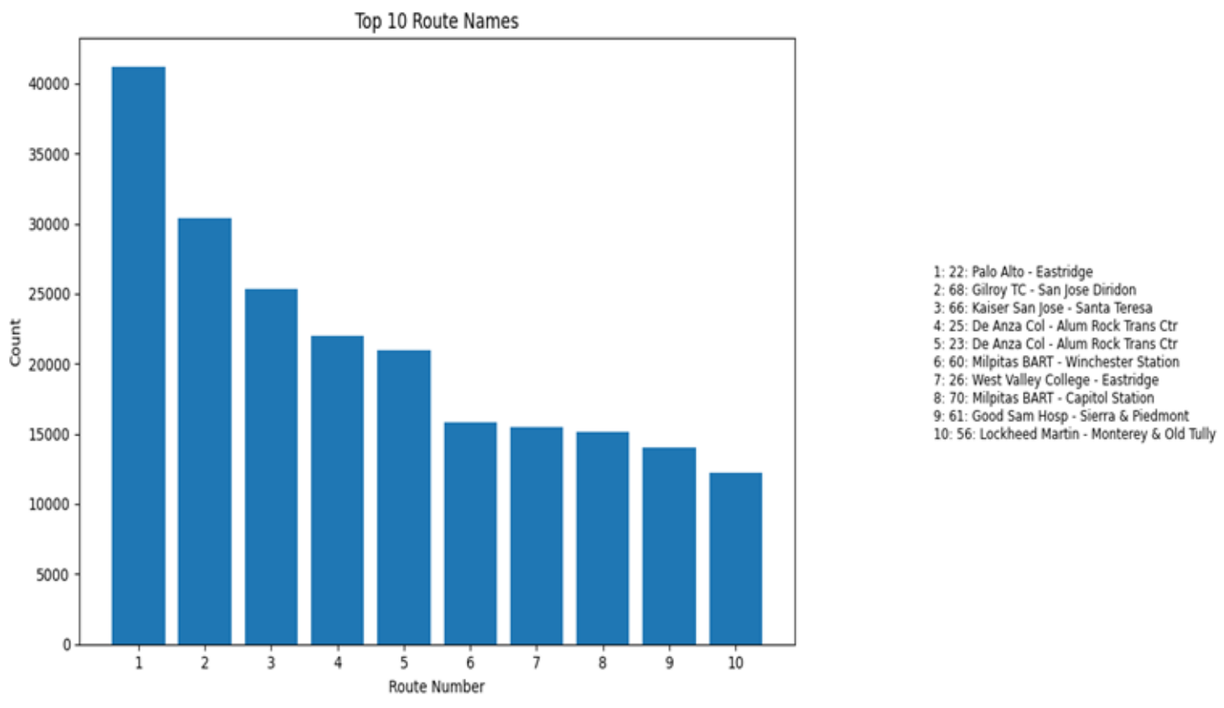
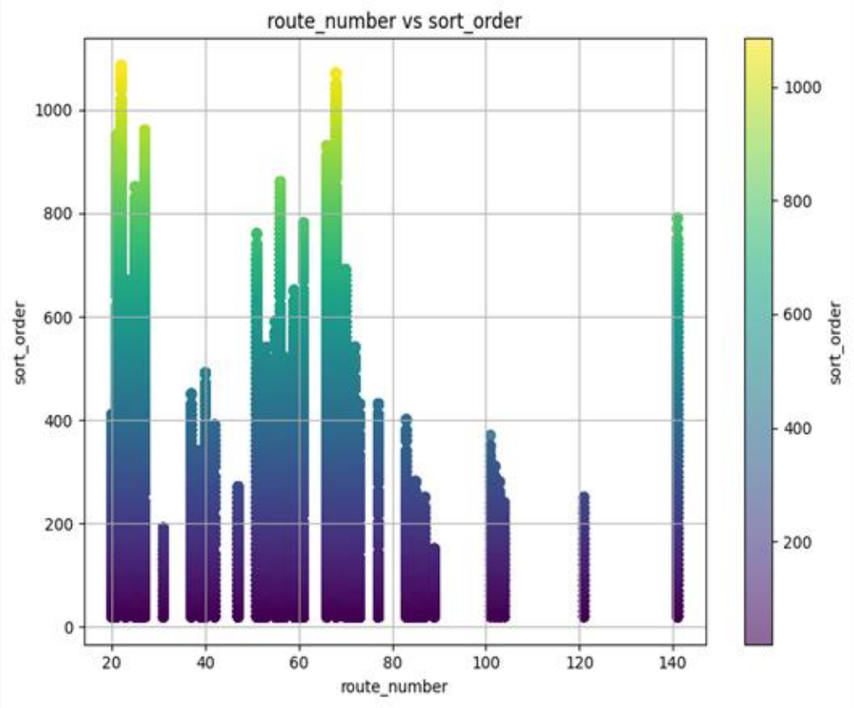
Exploratory Data Analysis (EDA) part 3





Bar chart Analysis

Route numbers and ridership levels relationship

Heat-Style Visualization

Color intensity showing route ridership patterns

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Interactive Dashboard - Tableau

View the live dashboard

Three Analytical Pillars

1. SPATIAL IMPACT

The Vibe: Identifying the network's "Gravity Centers."

The Data: Geographic Ridership Map locates city hubs. Top 10 Locations ranking pinpoints stops acting as the system's "Engine Room."

2. RIDERSHIP DENSITY

The Vibe: Measuring the "Intensity" of the commute.

The Data: Distribution Histogram visualizes "Thickness of Demand" - quiet suburban whisper vs high-volume downtown shout.

3. NETWORK FLOW

The Vibe: Tracking Silicon Valley's "Metabolic Rate."

The Data: Metric Toggle switches between Boardings (Inhale) and Alightings (Exhale) - reveals "Wellsprings" (Origins) vs "Anchors" (Destinations).

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Real-Time Diagnostic

Global Map Action

Clicking a city bubble filters all 9 charts instantly, maintaining context across views.

Metric Parameter

One-click toggle between Boardings and Alightings reveals directional flow patterns.

Reactive Sync

All KPIs and visualizations update in real-time, providing instant feedback to every interaction.

Summary: This dashboard provides a real-time diagnostic of the VTA network. By toggling between boardings and alightings and filtering by city, we can pinpoint exactly where the network is under the most pressure.

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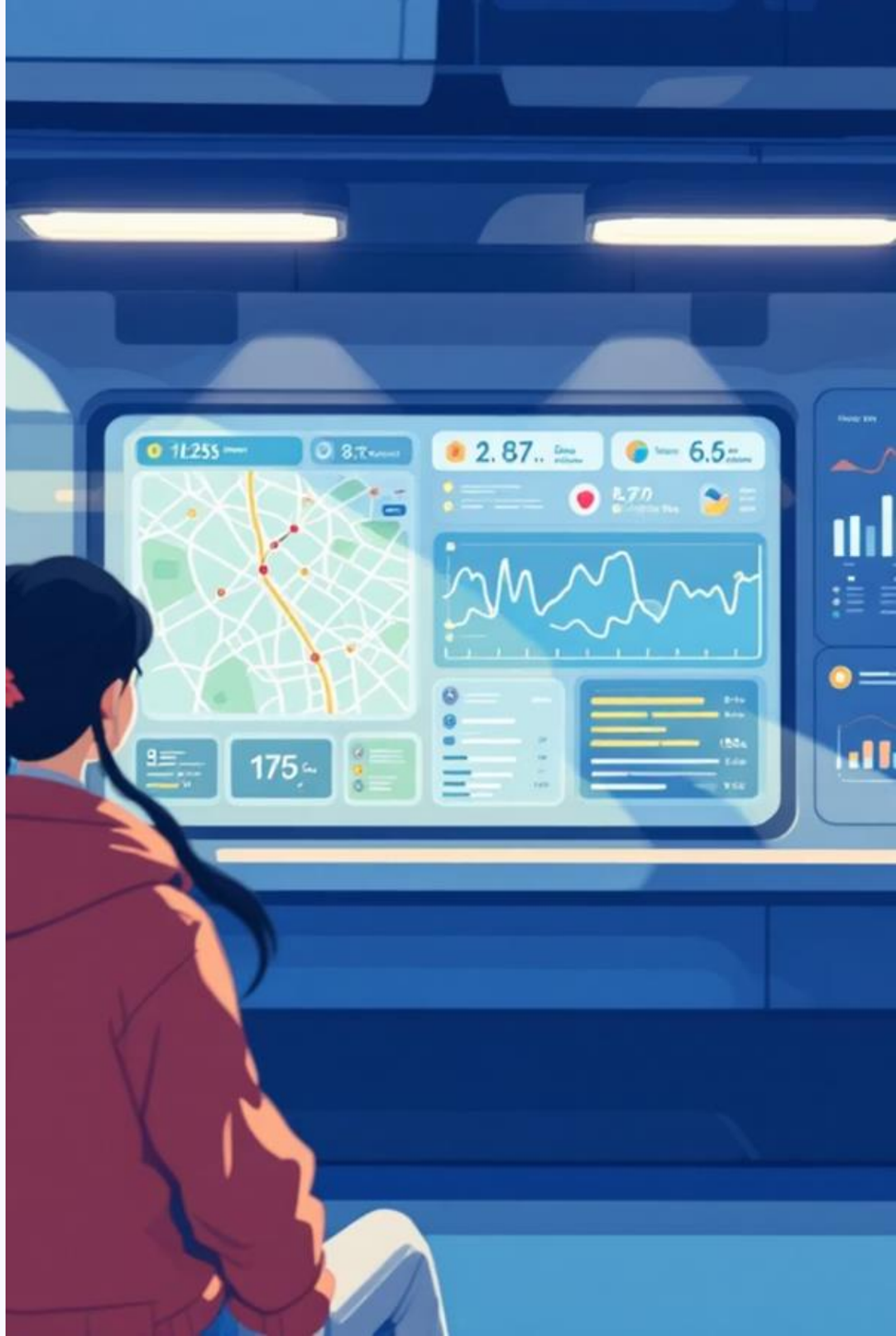
Interactive Dashboard - Plotly Dash

Story: "Where, When & How Silicon Valley Rides"

Analyze VTA ridership patterns across three distinct perspectives, powered by dynamic filters and real-time updates.



Three Analytical Perspectives

WHERE

Top 20 busiest stops, city ridership treemap, and top 15 routes by boardings reveal geographic hotspots.

WHEN

Hourly ridership heatmap, boardings by time period, and weekday vs weekend comparisons track temporal patterns.

HOW

Route type bubble chart, boarding vs alighting balance, and top 20 most crowded routes expose system dynamics.

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Dashboard Architecture

Dynamic KPI Cards

- Total Boardings | Unique Stops
- Routes Served | Cities Covered
- All metrics update with each filter change

Interactive Filters

- City Dropdown (global filter)
- Route Type Checklist: Express, Frequent, Local, Light Rail, Rapid, Other
- Service Period Checklist: Weekday, Saturday, Sunday

Visual Components

- 9 total visualizations across 3 tabs
- Horizontal bars, treemap, heatmap
- Grouped bars, bubble chart, scatter plot

Backend Engine

- 4 callbacks manage all updates
- Reactive architecture ensures instant response
- Plotly Dash framework

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WHERE Tab: Geographic Insights

01

Top 20 Busiest Stops

Horizontal bar chart identifies high-traffic boarding locations across the network.

02

City Ridership Treemap

Hierarchical visualization reveals which municipalities generate the most demand.

03

Top 15 Routes by Boardings

Ranking exposes the most heavily utilized corridors in the system.

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VTA 2025 Ridership Dashboard

Optimizing Silicon Valley Transit: Where, When & How People Ride

Filter by City: All Cities

Filter by Route Type: ☑ Express ☑ Frequent ☑ Light Rail ☑ Local ☑ Other ☑ Rapid ☑ Unknown

Total Boardings	Unique Stops	Routes Served	Cities Covered
1,704,646	3,119	60	21

WHERE: Stops & Cities | WHEN: Time Patterns | HOW: Performance & Crowding

Which stops, routes, and cities generate the most ridership?

Fig 1: Top 20 Busiest Stops — All Cities

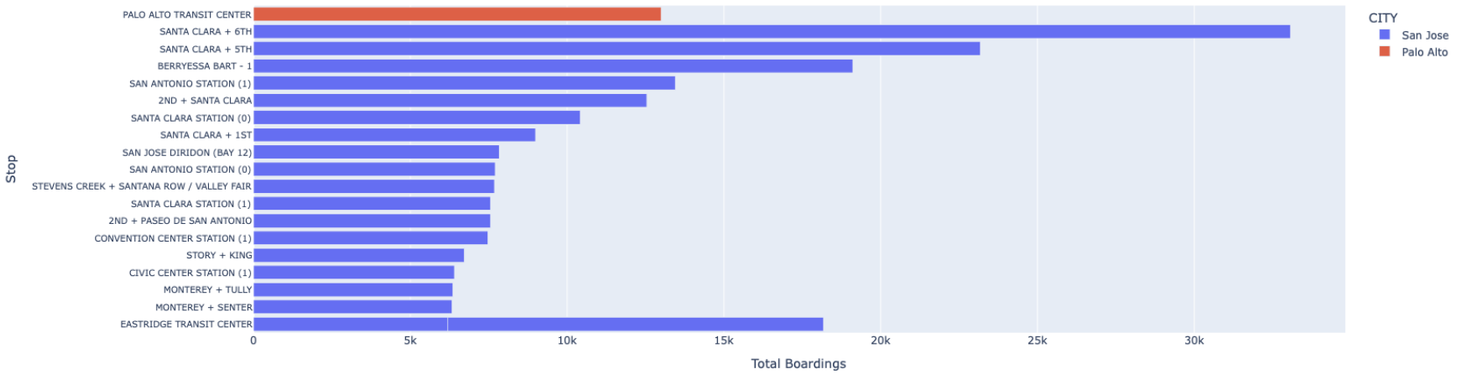


Fig 2: Ridership Distribution by City (Top 15)

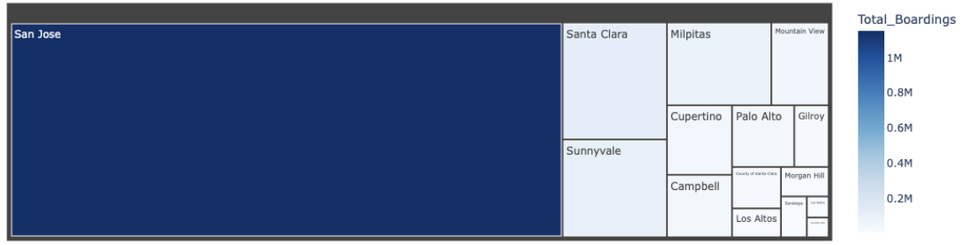
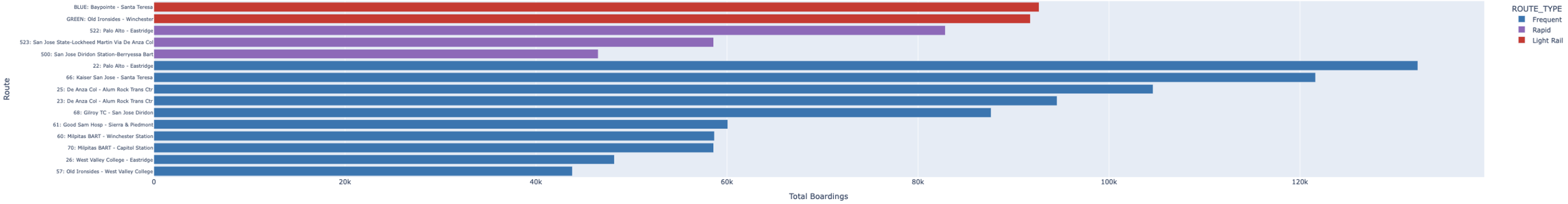


Fig 3: Top 15 Routes — All Cities



WHEN Tab: Temporal Patterns

Hourly Ridership Heatmap

Color-coded grid reveals peak hours and demand valleys throughout the day.

Boardings by Time & Route

Grouped bar charts compare ridership across route types during different service periods.

Weekday vs Weekend

Side-by-side comparison exposes how service patterns shift between work and leisure days.

Service Period Filter

Toggle between Weekday, Saturday, and Sunday to analyze demand profiles.

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VTA 2025 Ridership Dashboard

Optimizing Silicon Valley Transit: Where, When & How People Ride

Filter by City:
All Cities

Filter by Route Type:
☑ Express ☑ Frequent ☑ Light Rail ☑ Local ☑ Other ☑ Rapid ☑ Unknown

Total Boardings	Unique Stops	Routes Served	Cities Covered
1,704,646	3,119	60	21

WHERE: Stops & Cities | WHEN: Time Patterns | HOW: Performance & Crowding

When do riders use VTA the most - by hour, time period, and day type?

Filter by Service Period:
☑ Saturday ☑ Sunday ☑ Weekday

Fig 4: Ridership Heatmap — Hour of Day vs Service Period

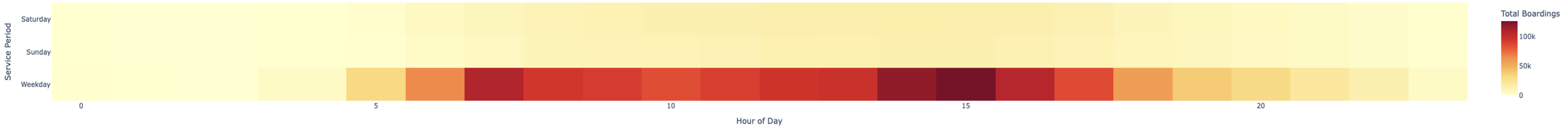


Fig 5: Boardings by Time Period & Route Type

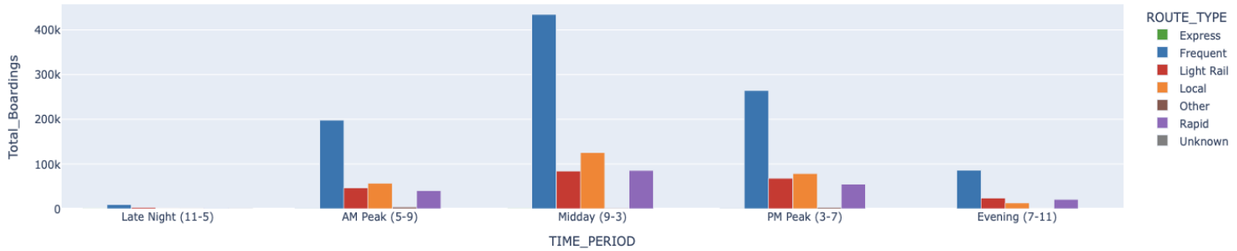
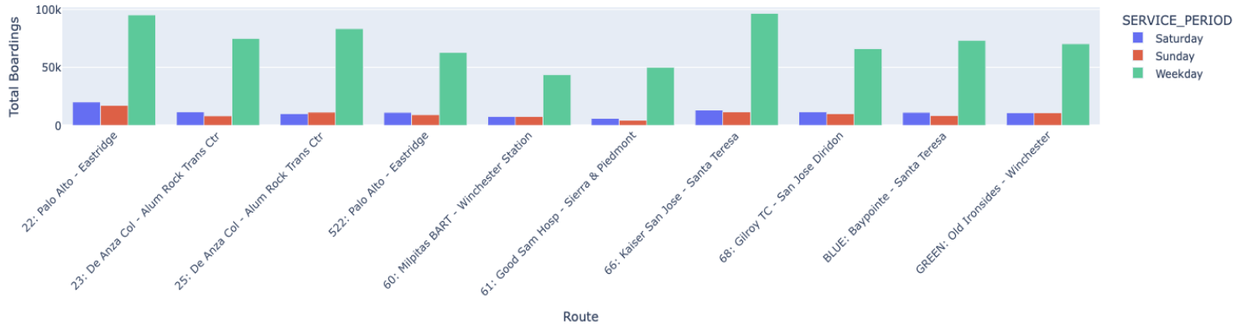


Fig 6: Weekday vs Weekend Ridership — Top 10 Routes



Data: VTA Oct 2025 Ridership by Stop & Station | Dashboard: Plotly Dash (CPU)

HOW Tab: System Dynamics

Route Type Bubble Chart

Size represents boardings, color encodes route type, position shows boarding vs alighting balance.

Boarding vs Alighting Balance

Scatter plot identifies routes that are net origins (more boardings) versus destinations (more alightings).

Top 20 Most Crowded Routes

Critical capacity metrics highlight where service improvements are most urgently needed.

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VTA 2025 Ridership Dashboard

Optimizing Silicon Valley Transit: Where, When & How People Ride

Filter by City: All Cities

Filter by Route Type: ☑ Express ☑ Frequent ☑ Light Rail ☑ Local ☑ Other ☑ Rapid ☑ Unknown

Total Boardings	Unique Stops	Routes Served	Cities Covered
1,704,646	3,119	60	21

WHERE: Stops & Cities | WHEN: Time Patterns | HOW: Performance & Crowding

How do different route types perform, and where is overcrowding or underutilization?

Fig 7: Route Type Performance (bubble = Avg Peak Load)

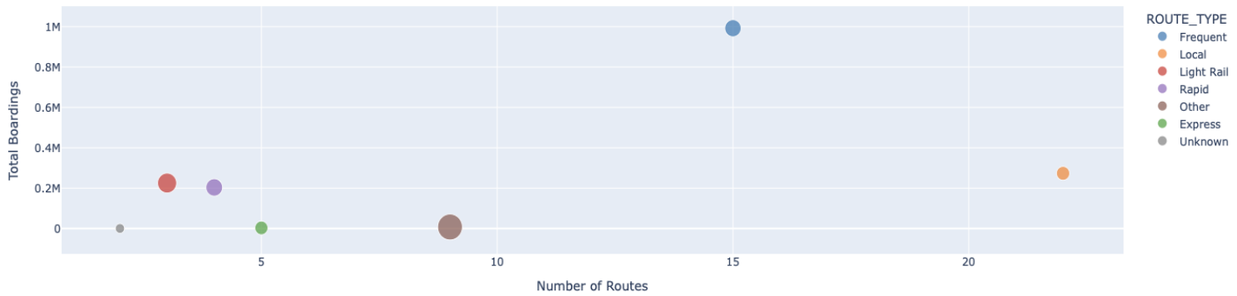


Fig 8: Boarding vs Alighting Balance by Route

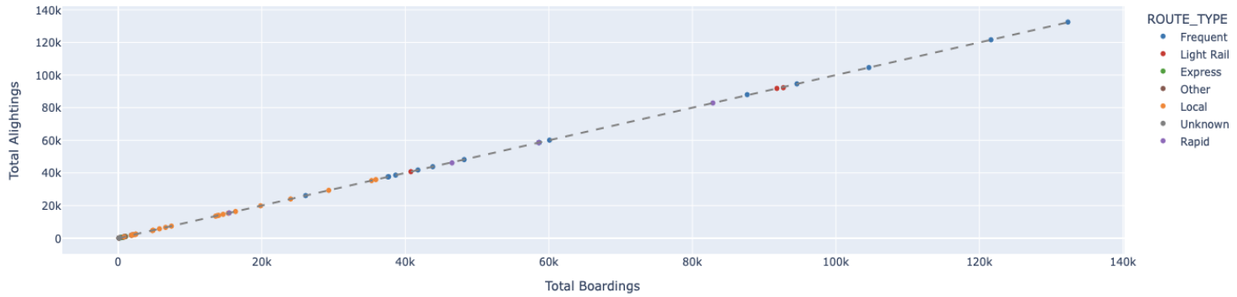
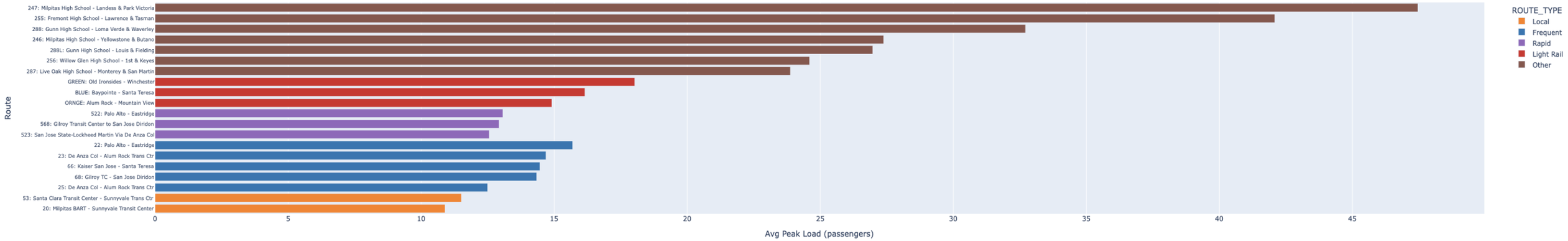


Fig 9: Top 20 Most Crowded Routes (Avg Peak Load)



Data: VTA Oct 2025 Ridership by Stop & Station | Dashboard: Plotly Dash (CPU)

GPU Acceleration with NVIDIA RAPIDS

Transforming Data Processing Speed

Load Excel
Read .xlsx with pandas (cuDF limitation)

Transfer to GPU
cudf.DataFrame(df_pd) moves data

GPU Processing
Run cleaning & aggregation with cuDF's pandas-like API

Render Dashboard
gdf.to_pandas() for Plotly

Why GPU Acceleration Matters

Fraction of the Time
Achieve massive speedups compared to traditional CPU-bound workflows.

Parallel Processing
Leverage thousands of cores to handle complex data calculations simultaneously.

Real-Time Interactivity
Enable fluid, responsive data exploration even on massive datasets.

Same Code Logic
Utilize familiar pandas-like APIs with minimal code refactoring.

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VTA 2025 Ridership Dashboard (GPU-Accelerated)

Optimizing Silicon Valley Transit: Where, When & How People Ride

Powered by NVIDIA RAPIDS cuDF

Filter by City:

All Cities

Filter by Route Type:

☑ Express ☑ Frequent ☑ Light Rail ☑ Local ☑ Other ☑ Rapid

Total Boardings	Unique Stops	Routes Served	Cities Covered
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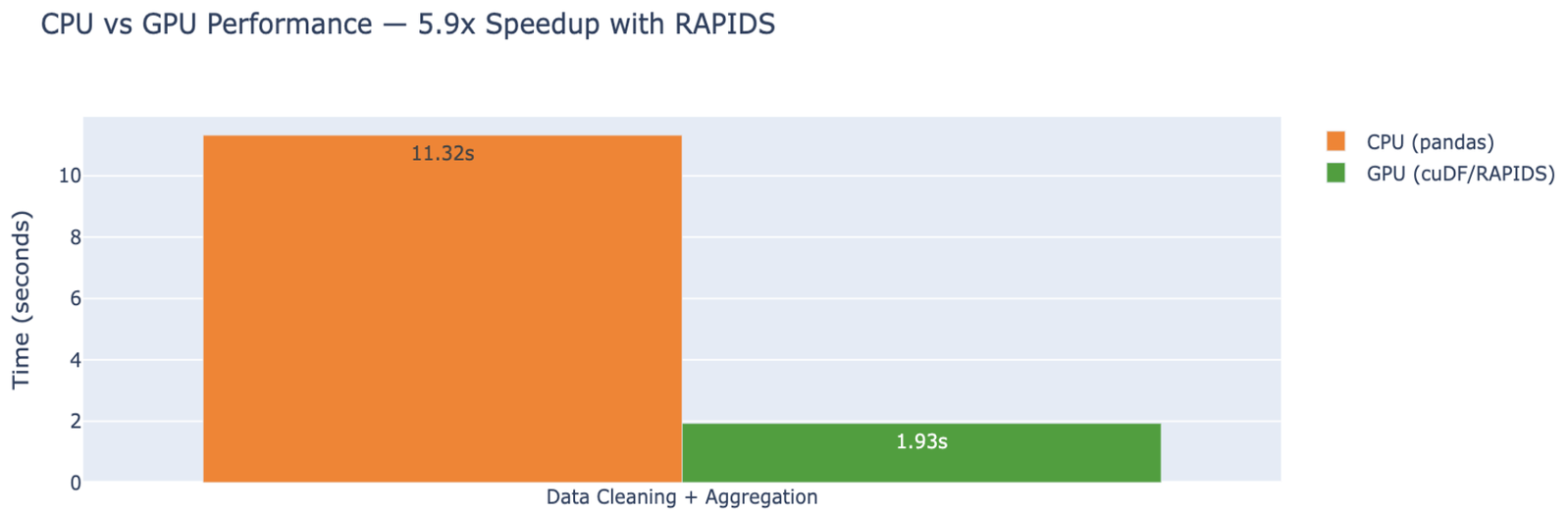
WHERE: Stops & Cities | WHEN: Time Patterns | HOW: Performance & Crowding | GPU vs CPU Benchmark

NVIDIA RAPIDS Acceleration Results

Dataset: 388K+ rows, 32 columns | CPU time: 11.32s | GPU time: 1.93s | Speedup: 5.9x



Preliminary ML Direction

To enhance urban mobility, we propose to leverage machine learning for precise transit ridership prediction. Accurate forecasts will be crucial for optimizing resource allocation, improving service efficiency, and better serving our community's transportation needs.

Classification: Predicting Route Crowding Levels

This proposed classification task will predict route crowding—categorized as Low, Medium, or High—to improve service planning and passenger experience. The target variable will be derived by binning the **PEAK_LOAD** metric using data-driven percentiles.

Key Features

The model will utilize essential predictors including **ROUTE_TYPE**, **TIME_PERIOD**, **SERVICE_PERIOD**, **CITY**, **TOTAL_ACTIVITY**, and **TRIPS** to capture temporal and operational dynamics.

Candidate Models

We will evaluate performance across three robust classification algorithms:

- Logistic Regression
- Random Forest
- XGBoost

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Regression: Predicting Boardings per Stop

The second proposed task will address the regression of **BOARDINGS**–a continuous numeric target–based on temporal and route-level features identified through exploratory data analysis to accurately forecast future passenger demand.

Key Features

The model will utilize predictive features including **ROUTE_TYPE**, **ROUTE_NUMBER**, **TIME_PERIOD**, **SERVICE_PERIOD**, **CITY**, and **TRIP_HOUR** to capture spatial and temporal patterns.

Candidate Models

We propose to evaluate performance across three robust regression algorithms:

- Linear Regression
- Random Forest Regressor
- XGBoost

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Next Steps

01

Feature Engineering

Create time-based features (hour bins, peak/off-peak flags)

02

Model Training

Train candidate models with cross-validation

03

Evaluation

Assess performance metrics (accuracy, precision, recall for classification; RMSE, MAE for regression)

04

Deployment

Integrate best-performing models into ridership forecasting pipeline

GitHub Repository

Team Member	Contribution
Sneha Tumkur Narendra	Developed Plotly Dash dashboard using CPU and RAPIDS (NVIDIA GPU acceleration) for performance optimization and scalable analytics.
Sadaf Fatima Syeda	Designed and implemented Tableau dashboards, including geographic maps, ridership distributions, and KPI summaries.
Naeem Mannan	Conducted Exploratory Data Analysis (EDA), including data cleaning, preprocessing, and identifying key patterns and insights from the dataset.

View on GitHub

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Thank You!

Questions?

We welcome your thoughts, feedback, and any questions you may have about our preliminary findings and future directions in enhancing urban mobility.